Puma Marketing Strategies and Parkruns

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Overview

Key Insights

Question One: Which Parkrun provides the best marketing opportunity with the largest number of people?

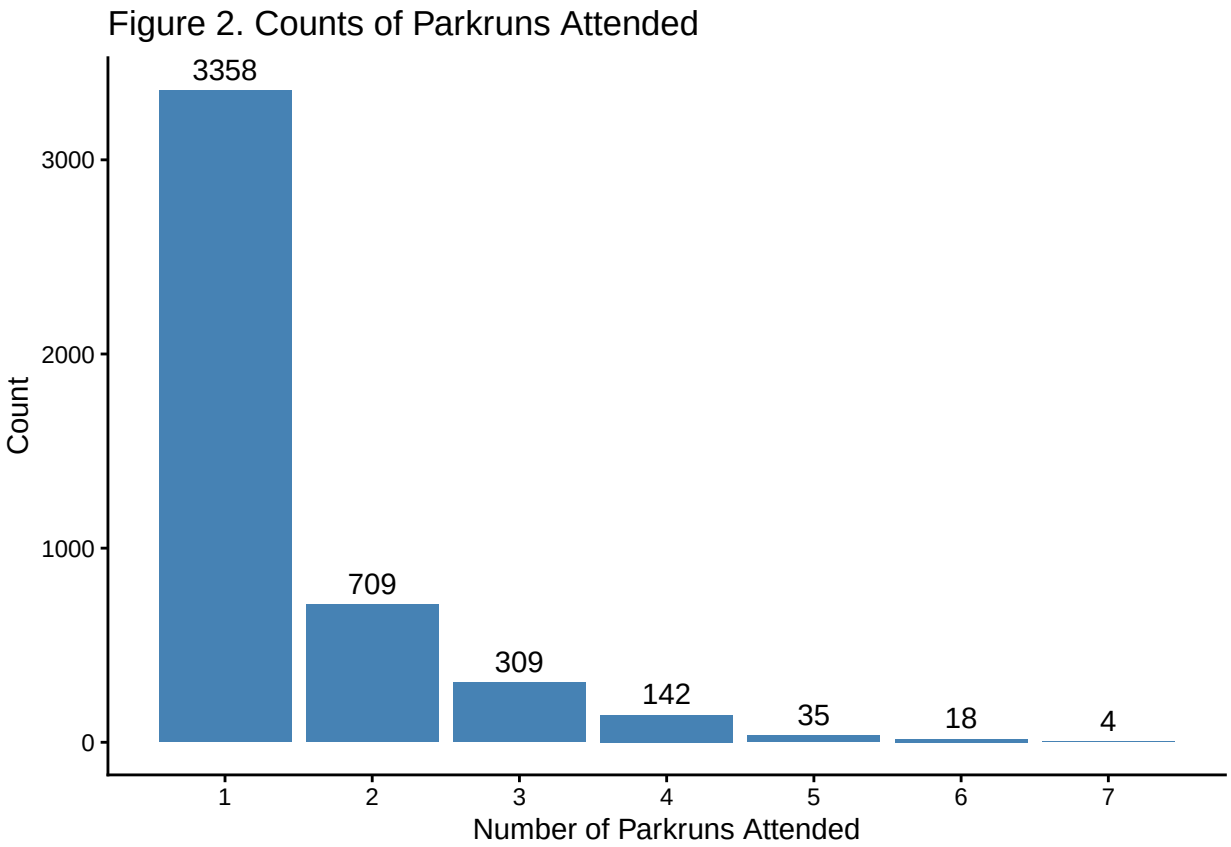
Question Two: Do new people come each week or is it always the same people?

Question Three: Do people move between Parkruns? How does this affect advertising?

In order to determine whether or not runners attend different Parkruns, we can simply count the number of Parkrun locations each runner has been to in the past 11 weeks. However, due to unidentifiability, we cannot include any unknown runners in this question.

Table 1: Proportion of runners who have attended a different parkrun

Total Runners	Mixed Runners	Percentage of Mixed Runners
4575	1217	26.6



The above table (Figure 1) shows that there is a significant number of runners who have attended Parkruns at mulitple locations (26.6%), or about 1200 runners out of 4600. However, Figure 2 shows that the vast majority of runners have only attended events at one Parkrun location, with very few reaching out to others.

However, this 26% overlap means that advertising across all locations is not going to be particularly budget effective, as advertisements will end up being displayed to people who have already bought Puma products. Therefore, advertising across fewer Parkrun locations can be a more economical option, as it would allow for the same effective advertisement reach, at a lower cost when compared to advertising at all locations.

Question Four: Where do elite runners like to run?

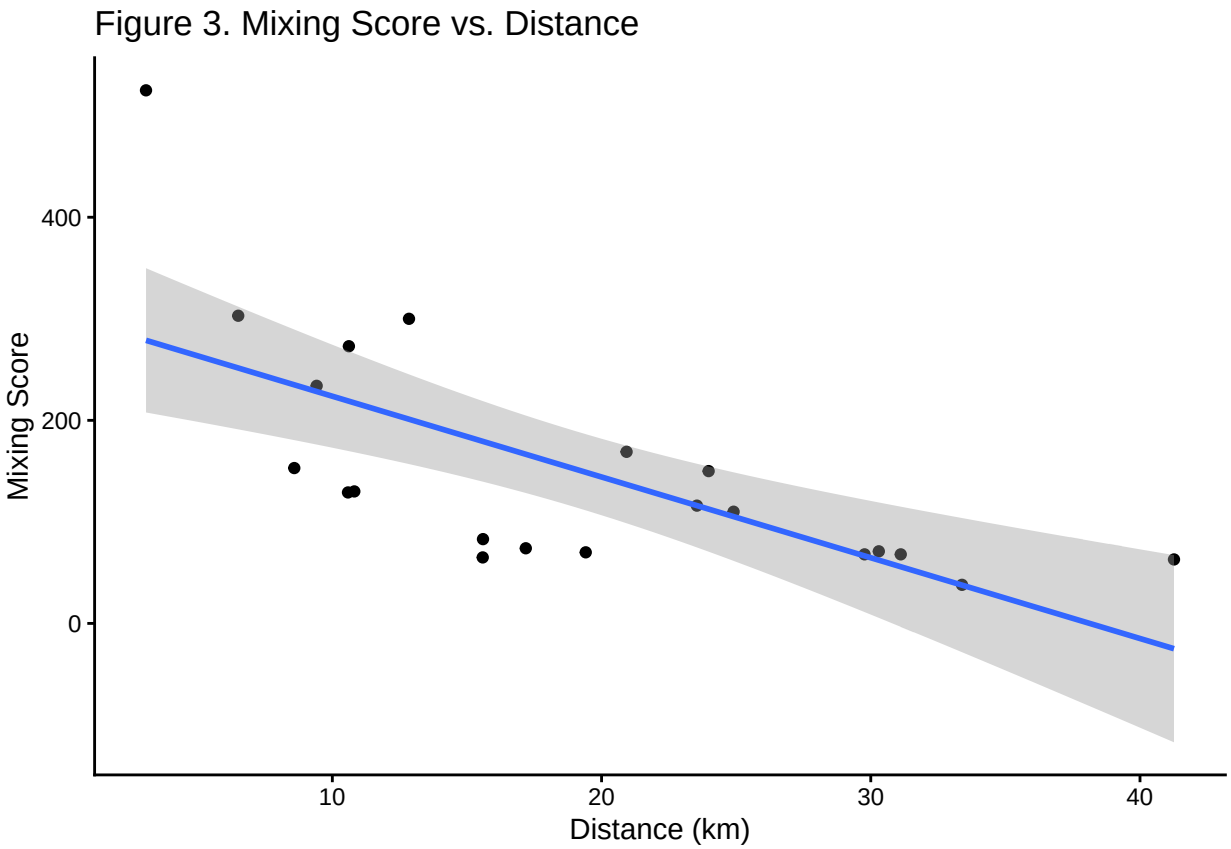
Question Five: What locations feature the most consistent runners?

Question Six: Does distance between Parkruns affect how much mixing occurs?

In order to determine whether or not the distance between two Parkruns predicts how much mixing occurs, we can obtain each unique pair of Parkruns, and then using geographical data, determine the distances between these pairs, effectively creating a map of each Parkrun location.

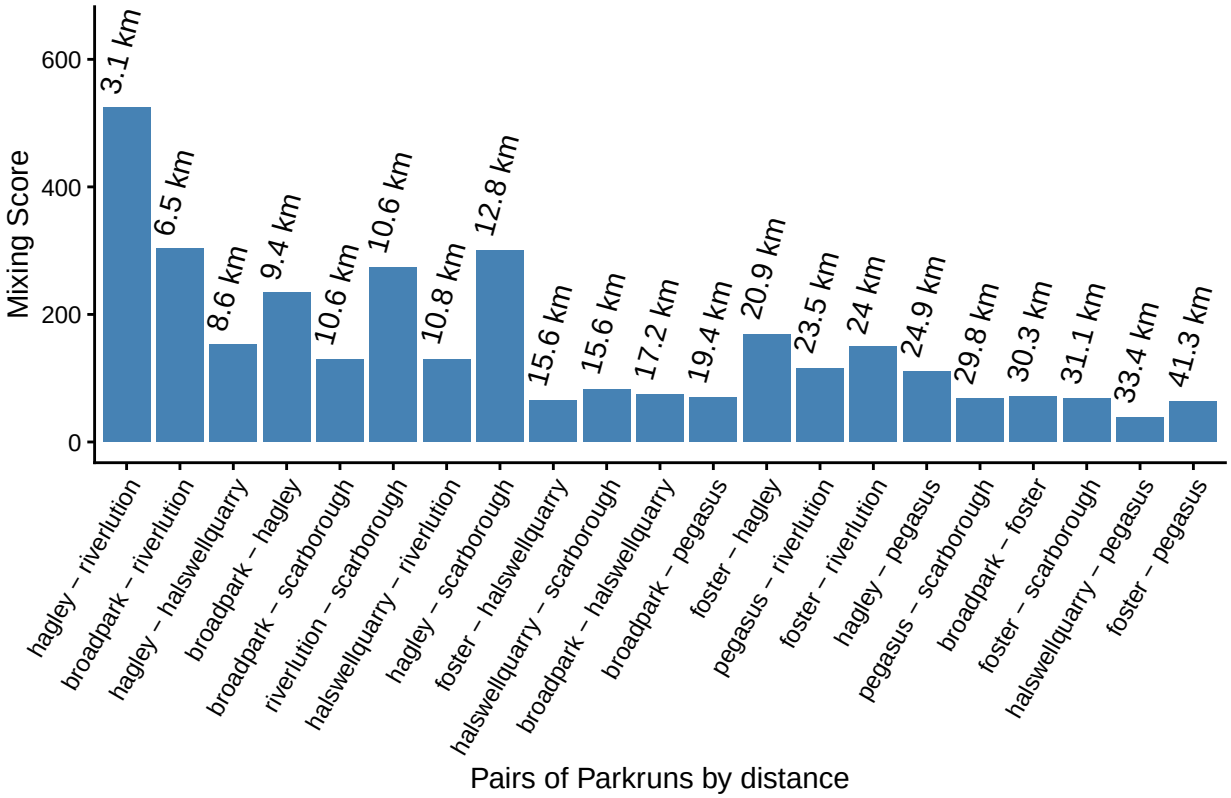
We can then count the number of runners who have gone to both locations in these pairs, and set that as our Mixing Score. A higher Mixing Score means more runners have attended both Parkruns in the past 11 weeks.

```
## `geom_smooth()` using formula = 'y ~ x'
```



As seen above, Figure 3 shows that there is a negative correlation between Distance and Mixing Score. In other words, as the distance between parkruns increases, the amount of mixing decreases.

Figure 4. Parkrun Pairs and Mixing Scores arranged by distance



The above bar chart (Figure 4), also shows this correlation, but also labels each parkrun pair with the distance between them. As is seen in the chart, the Hagley and Riverlution parkruns are very close (only 3km apart), and as such, receive a very large amount of mixing.

Summary and Conclusion